



**Panchip Microelectronics Co., Ltd.**

## **PAN3029 Crystal Selection Guide**

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## Document Description

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# 1 Introduction

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## 1.1 Background Overview

PAN3029 is a wireless transceiver chip used for long-distance communication. The chip has a minimum sensitivity of -143dBm, a maximum transmit power of 20dBm, and an industry-leading link budget of 163dB. At a transmit power of 20dBm, the chip generates a lot of heat, which is transferred to the crystal through the PCB. Since the frequency of the crystal changes with temperature, frequency drift will occur during the transmission process. This drift will pose a great challenge to the receiving demodulation. In severe cases, it will lead to receiving demodulation data errors and packet loss.

## 2 Frequency drift mechanism and influencing factors

Crystal frequency drift is mainly related to the temperature characteristics of the crystal itself and the temperature change. The crystal temperature change depends on the heat dissipation of the chip and the heat change from the chip to the crystal through the PCB. The heat dissipation of the chip is mainly related to the transmission power, while the heat change from the chip to the crystal is mainly related to the PCB size and the heat insulation between the crystal and the chip.

The reception tolerance to frequency drift is related to BW, SF, LDR, Payload length and working frequency band. For different configurations, each symbol has different requirements for frequency drift, and the total transmission time of a frame is also different, resulting in different requirements for frequency drift.

The following is a detailed description of each influencing factor.

### 2.1 Power dissipation

The following is the power dissipation at 20dBm, 14dBm, and 10dBm in the 490MHz frequency band. The calculation formula for power dissipation is: power supply voltage VCC, chip current is I, input power is Pin, output power is Pout, and power dissipation is Pd.

$$Pd = Pin - Pout = VCC * I - Pout$$

| VCC (V) | I (mA) | Pin (mW) | Pout (dBm) | Pout (mW) | Pd (mW) |
|---------|--------|----------|------------|-----------|---------|
| 3.3     | 91     | 300.3    | 20         | 100       | 200.3   |
| 3.3     | 50     | 165      | 14         | 25.1      | 139.8   |
| 3.3     | 30     | 99       | 10         | 10        | 89      |

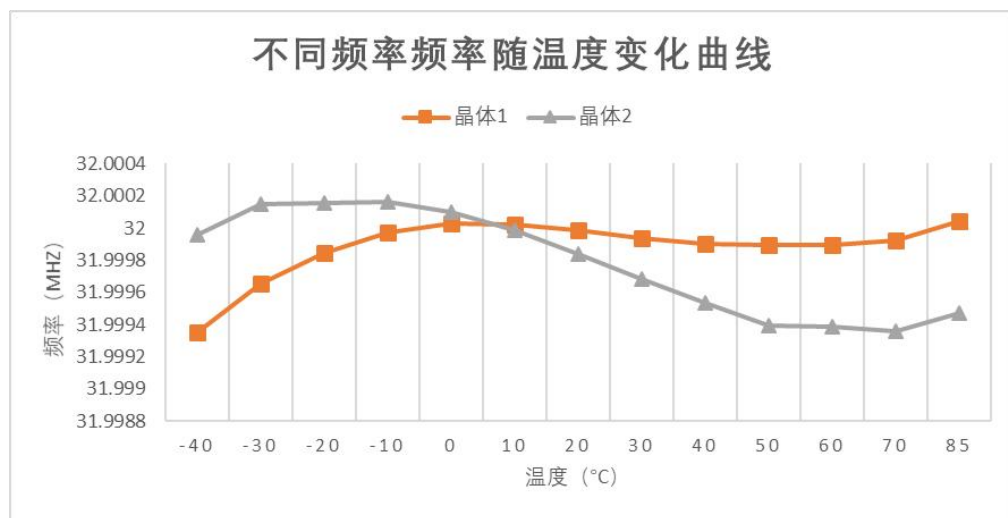
From the table above, we can see that 10dBm will reduce the power dissipation by more than half compared to 20dBm.

### 2.2 Crystal frequency characteristics

The following tests are conducted on 32M crystals from different manufacturers. By observing the change of single carrier frequency over time in the 490MHz band and counting the drift within a certain period of time after the single carrier is launched, it can be seen that the drift of different crystals varies greatly. The ones with smaller frequency drift are 1C232000AA0B from Big Vacuum and FSX-3M from Fuji.

| 厂家   | 料号                        | 立创编号      | 封装形式        | 负载电容 (pF) | 温度特性 (-20~70°) ppm | 频率精度 (ppm) | 频率漂移 (Hz) |
|------|---------------------------|-----------|-------------|-----------|--------------------|------------|-----------|
| 大真空  | 1C232000AA0B              | /         | SMD3225-4P  | 12        | 10                 | 10         | 250       |
| 泰晶   | SX32Y032000B91T           | C271639   | SMD3225-4P  | 9         | 10                 | 10         | 342       |
| 泰晶   | SX32Y032000BC1T           | C271658   | SMD3225-4P  | 12        | 10                 | 10         | 496       |
| 安基   | CXA-032000-3F7E40         | C369204   | SMD3225-4P  | 12        | 15                 | 10         | 800       |
| 晶技   | 7M32000020                | C5203632  | SMD3225-4P  | 12        | /                  | /          | 738       |
| 爱普生  | X1E000021016900           | C255932   | SMD3225-4P  | 12        | 10                 | 10         | 488       |
| 晶光华  | S3232000091040            | C426986   | SMD3225-4P  | 12        | 30                 | 30         | 460       |
| 晶科鑫  | 7U32000E12UCG             | C251626   | SMD3225-4P  | 12        | 10                 | 10         | 528       |
| 川晶   | C3SB32.000F12E12          | C1122385  | SMD3225-4P  | 12        | 20                 | 10         | 500       |
| 长晶   | CJ13-320001210A20         | C712874   | SMD3225-4P  | 12        | 20                 | 10         | 490       |
| 晶优   | JYXT32S4-032.00000-93C4B0 | C357891   | SMD3225-4P  | 12        | 20                 | 10         | 760       |
| 浩都   | 7B032000M01               | C655025   | SMD3225-4P  | 12        | 20                 | 10         | 384       |
| 台湾加高 | X3S032000B91H-HU          | C254375   | SMD3225-4P  | 9         | 20                 | 10         | 716       |
| 晶威特  | CM4032M00012T8188141      | C2932987  | SMD3225-4P  | 12        | 15                 | 10         | 330       |
| 晶威特  | CF4032M00012T5074007      | C16197273 | SMD3225-4P  | 12        | 20                 | 10         | 292       |
| 晶威特  | CF4032M00012T5074007      | C5765975  | SMD3225-4P  | 12        | 20                 | 10         | 850       |
| 晶宝   | 01X.MD.110HJV0032000000   | C7570525  | SMD3225-4P  | 10        | 10                 | 10         | 380       |
| 富士   | FSX-3M                    | /         | SMD3225-4P  | 12        | /                  | 10         | 224       |
| 晶威特  | CN4032M00009T8188116      | C2932962  | SMD 2016-4P | 9         | 30                 | 10         | 522       |
| 晶威特  | CN4032M00012T8188117      | C2932963  | SMD 2016-4P | 12        | 15                 | 10         | 516       |
| 鸿星   | E3SB32E00000BE            | C5241172  | SMD3225-4P  | 10        | 10                 | 10         | 470       |
| 索斯特  | 2520+32M+9PF+10PPM        | C5181488  | SMD3225-4P  | 12        | /                  | 10         | 670       |

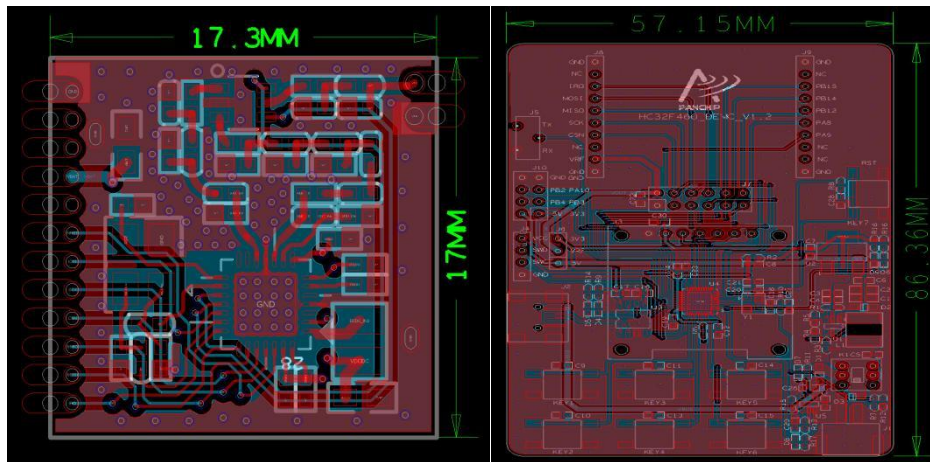
The figure below shows the temperature characteristics of two crystals. The frequency drift of crystal 1 is 250Hz, and the frequency drift of crystal 2 is 850Hz. It can be seen from the curve that the frequency characteristic stability of crystal 1 is better in the range of -10~70°C, while that of crystal 2 is worse. The test results of temperature characteristics and frequency drift are consistent.



## 2.3 PCB design drift impact

Comparing the frequency drift of the same crystal on two PCBs with different sizes, it is found that the larger the PCB, the better the heat dissipation and the smaller the frequency drift. Under the same time, the small module drifts to 850Hz, while the R&D EVB test board drifts to only 288Hz.

This shows that the larger the board, the better the heat dissipation and the smaller the impact on the crystal temperature. In addition to size, it is also important to do a good job of PCB thermal isolation of the crystal and chip, which needs to be considered in PCB thermal design.



## 2.4 Transmission time

For different BW and SF, the duration of each symbol is different, and the total packet transmission time will also be different, which can be calculated using the PAN3029 chip calculator.

| PAN3029芯片计算器              |        |      |                            |
|---------------------------|--------|------|----------------------------|
| 输入参数                      | 值      | 单位   | 描述                         |
| 有效负载字节 (2 to 255)         | 64     | Byte | 用户有效负载长度                   |
| 带宽                        | 125.0  | KHz  | -                          |
| SF因子                      | 5      | -    | -                          |
| 载波频率 (150 MHz to 1.2 GHz) | 470    | MHz  | -                          |
| 噪声系数                      | 6      | dB   | 默认噪声系数为6                   |
| 发射输出功率 (-7 to +22)        | 22.0   | dBm  | 辐射发射功率                     |
| 报文模式                      | 1      | -    | 默认支持报文                     |
| 码率                        | 1      | -    | 1=4/5, 2=4/6, 3=4/7, 4=4/8 |
| 前导码符号数里                   | 8      | -    | 默认值为8, 输入范围8~65535         |
| 低速率模式                     | 0      | -    | 1=支持低速率模式, 0=不支持低速率模式      |
| 输出参数                      | 值      | 单位   | 描述                         |
| 接收灵敏度                     | -119.5 | dBm  | -                          |
| 链路预算                      | 142    | dB   | -                          |
| 符号长度                      | 0.3    | ms   | -                          |
| 前导码持续时间                   | 3.6    | ms   | -                          |
| 有效载荷持续时间                  | 36.6   | ms   | -                          |
| 总发包时间                     | 40.3   | ms   | 包含CRC16                    |
| 实际数据速率                    | 12.72  | kbps | 包含前导码/报头/CRC               |

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The following lists and calculates the symbol time and total time for sending a frame for different SF and BW. The smaller the BW, the larger the SF, the longer the symbol time, the longer the total time for a frame, the greater the frequency drift, and the higher the requirement for frequency stability.

| BW   | SF | CR | Preamble数量 (bit) | Payload数量 (Bytes) | 符号时间 (ms) | 一帧总时间 (ms) |
|------|----|----|------------------|-------------------|-----------|------------|
| 125  | 5  | 1  | 20               | 64                | 0.3       | 40.3       |
| 125  | 6  | 1  | 20               | 64                | 0.5       | 67.7       |
| 125  | 7  | 1  | 20               | 64                | 1         | 118        |
| 125  | 8  | 1  | 20               | 64                | 2         | 215.6      |
| 125  | 9  | 1  | 20               | 64                | 4.1       | 390.1      |
| 125  | 10 | 1  | 20               | 64                | 8.2       | 698.4      |
| 125  | 11 | 1  | 20               | 64                | 16.4      | 1314.8     |
| 500  | 12 | 1  | 20               | 64                | 8.2       | 616.4      |
| 250  | 12 | 1  | 20               | 64                | 16.4      | 1232.9     |
| 125  | 12 | 1  | 20               | 64                | 32.8      | 2465.8     |
| 62.5 | 12 | 1  | 20               | 64                | 62.5      | 4931.6     |

## 2.5 Carrier frequency impact

The carrier frequency  $f_{TX}$  of TX end is  $f_{REFTX} \cdot (n_{TX} + m_{TX})$ , the carrier frequency  $f_{RX}$  of RX end is  $f_{REFRX} \cdot (n_{RX} + m_{RX})$ , the absolute value of the intermediate frequency  $f_{IF}$  of RX end is  $f_{TX} - f_{RX}$ . The frequency drift during the transmission process of TX end affects the frequency drift of the intermediate frequency of RX end. The magnitude of the frequency drift is related to the stability of the crystal  $f_{REFTX}$  of TX end and the multiple  $n_{TX} + m_{TX}$ . And  $n_{TX} + m_{TX}$  is related to the operating frequency, that is, the higher the frequency, the greater the impact of  $f_{REFTX}$  drift. For the same crystal, the same output power, assuming the power dissipation is the same, the frequency drift of 866M is twice that of 433M.



## 3 Crystal Recommendation

### 3.1 Passive Crystal

Based on Panqi's DTM-CC0015 module, the operating frequency band is 408~510MHz, the operating temperature is -20~70°C, the transmission payload is no more than 20Bytes, the transmission power is 20dBm, and the BW, SF, and LDR configurations are shown in the following table. Passive crystals with good temperature characteristics can be used, and the frequency accuracy is required to be 10ppm, and the temperature stability of -20~70°C is 10ppm. KDS's 1C232000AA0B

The test effect will be better than other passive crystals. If the cost is acceptable, it is recommended to use KDS's 1C232000AA0B.

| BW (KHz) | SF | LDR |
|----------|----|-----|
| 500      | 5  | OFF |
| 500      | 6  | OFF |
| 500      | 7  | OFF |
| 500      | 8  | OFF |
| 500      | 9  | OFF |
| 500      | 10 | OFF |
| 500      | 11 | OFF |
| 500      | 12 | OFF |
| 250      | 5  | OFF |
| 250      | 6  | OFF |
| 250      | 7  | OFF |
| 250      | 8  | OFF |
| 250      | 9  | OFF |
| 250      | 10 | OFF |
| 250      | 11 | OFF |
| 125      | 5  | OFF |
| 125      | 6  | OFF |
| 125      | 7  | OFF |
| 125      | 8  | OFF |
| 125      | 9  | OFF |
| 125      | 10 | OFF |
| 62.5     | 5  | OFF |
| 62.5     | 6  | OFF |
| 62.5     | 7  | OFF |
| 62.5     | 8  | OFF |
| 62.5     | 9  | OFF |

Based on the above conditions, changing the power, temperature, frequency band and payload length will change the BW and SF configuration of the available passive crystal. It may increase the available range by one gear at the lowest rate (BW decreases by one gear, or SF increases by one gear), or it may decrease the available range by one gear at the lowest rate (BW is too high by one gear, or SF decreases by one gear). If multiple conditions change, the effects of multiple gear changes will be superimposed. For example:

Note 1: When the power is reduced to about 10dBm, 125K SF11 can use passive crystals;

Note 2: When the operating temperature reaches -40°C, 125K SF10 and 62.5 SF9 cannot use passive crystals;

Note 3: When the operating frequency band is 863~870MHz and 902~928MHz, 125K SF10 and 62.5 SF9 cannot use passive crystals;

Note 4: When the Payload length is greater than 20Bytes and less than 70Bytes, 125K SF10 and 62.5 SF9 cannot use passive crystals.

## 3.2 Temperature Compensated Crystal Oscillator

Configurations that cannot use passive crystals require the use of 1ppm active temperature-compensated crystals, and all operating temperatures must meet the 1.5ppm requirement. For applications between -40 and 85°C, KDS's active temperature-compensated crystal 1XXD32000MBA is recommended.

( $T_A = -40 \sim +85^\circ\text{C}$ ,  $L_{\text{OAD\_R}}/C = 10\text{k}\Omega/10\text{pF}$ ,  $V_{\text{CC}} = +3.3\text{V}$ , unless otherwise noted)

|   | Item                | Conditions                                                                            | Limits |      |       | unit             | Notes |
|---|---------------------|---------------------------------------------------------------------------------------|--------|------|-------|------------------|-------|
|   |                     |                                                                                       | min.   | typ. | max.  |                  |       |
| 1 | Current Consumption |                                                                                       | -      | -    | +2.0  | mA               |       |
| 2 | Output Level        |                                                                                       | 0.8    | -    | -     | V <sub>P-P</sub> | 1     |
| 3 | Symmetry            | GND level (DC cut)                                                                    | 40/60  | -    | 60/40 | %                |       |
| 4 | Harmonics           |                                                                                       | -      | -    | -5    | dBc              |       |
| 5 | Frequency Stability |                                                                                       |        |      |       |                  |       |
|   | 1.Tolerance         | After 2 times reflow<br>Ref. to nominal frequency                                     | -      | -    | ±1.5  | ppm              | 2,3   |
|   | 2.vs Temperature    | $T_A = -30 \sim +85^\circ\text{C}$<br>Ref. to frequency ( $T_A = +25^\circ\text{C}$ ) | -      | -    | ±0.5  | ppm              |       |
|   |                     | $T_A = -40 \sim -30^\circ\text{C}$<br>Ref. to frequency ( $T_A = +25^\circ\text{C}$ ) | -      | -    | ±1.0  | ppm              |       |
|   | 3.vs Supply Voltage | $V_{\text{CC}} = +3.3\text{V} \pm 5\%$                                                | -      | -    | ±0.2  | ppm              |       |
|   | 4.vs Load Variation | $L_{\text{OAD\_R}}/C = (10\text{k}\Omega/10\text{pF}) \pm 10\%$                       | -      | -    | ±0.2  | ppm              |       |
|   | 5.vs Aging          | $T_A = \text{Room ambient}$                                                           | -      | -    | ±1.0  | ppm/year         |       |
| 6 | Start Up Time       | @90% of final V <sub>out</sub> level                                                  | -      | -    | 2.0   | ms               |       |
| 7 | SSB Phase Noise     | Relative to F0 level offset 1kHz                                                      | -      | -    | -125  | dBc/Hz           |       |